Automated Coin Counter
Project Interfaces and Verification-Validation Plan

PROCTECH 2SE3 – Systems Engineering Fundamentals

Winter 2026

The W Booth School of Engineering Practice and Technology

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Date of submission: March 22nd , 2026

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1. Project Description

Automated coin sorting and counting system is designed to recognize, sort, and count mixed coins without requiring human involvement. It is used in commercial and financial environments where efficiency and accuracy are important. The system reduces manual handling and operational errors while maintaining safe and reliable operation. It continuously monitors operating conditions and responds in a controlled manner. It is also designed to operate safely during power outages and to make routine maintenance easier.

2. Interfaces

The external interfaces of the Automated Coin Sorting and Counting System are categorized into four categories: physical, mass, energy, and information interfaces. Each interface is clearly described along with its purpose.

2.1 Physical Interfaces

1. Coin Input Hopper

Allows the user to input mixed Canadian coins into the system. It serves as the primary entry point for all coins.

2. Output Collection Bins

Stores sorted coins by denomination (nickel, dime, quarter, loonie, toonie) and allows users to retrieve them after processing.

3. Reject Tray

Collects invalid items such as foreign coins, damaged coins, or non-coin objects to prevent contamination of valid outputs.

4. User Control Buttons (Start/Stop/Emergency Stop)

Allows the operator to start, stop, or immediately stop the system for safe operation.

5. Machine Housing and Enclosure

Protects internal components and prevents users from coming into contact with moving parts during operation.

6. Access Panel and Maintenance Cover

Provides access for maintenance, inspection, and cleaning of internal components.

2.2 Mass Interfaces

1. Mixed Coin Input Flow

Represents the flow of mixed coins entering the system through the input hopper.

2. Sorted Coin Outputs

Represents the separated coin streams exiting the system into individual bins for each denomination.

3. Reject Coin Flow

Represents the flow of invalid or rejected coins directed into the reject tray.

4. Overflow Coin Flow

Represents coins that are redirected when a bin reaches full capacity.

2.3 Energy Interfaces

1. AC Power Input (120V)

Provides electrical power from an external source required for system operation.

2. Internal Power Supply Interface

Converts and distributes electrical energy to subsystems such as sensors, controller, and actuators.

3. Motor Power Interface

Supplies electrical energy to motors responsible for coin transport and sorting mechanisms.

4. Sensor Power Interface

Provides electrical energy to detection sensors used for coin identification and counting.

2.4 Information Interfaces

1. Sensor to Controller Signals

Transfers detection data such as coin presence, size and type from sensors to the control system.

2. Controller to Actuator Signals

Sends control commands to motors and sorting mechanisms to ensure coins are routed correctly.

3. Count Display Interface

Displays the total value and the count for each denomination to the user after processing.

4. System Status Indicators (LED and Display)

Communicates system states such as ready, running, jam, fault, and maintenance required.

5. User Input Commands

Transfers user commands such as start, stop, reset into the control system.

6. Fault and Jam Notification Interface

Provides alerts to the user when abnormal conditions such as jams or system failures occur.

3. Verification and Validation Plan

The following verification and validation processes ensure that the Automated Coin Sorting and Counting System meets all functional, performance, and safety requirements.

1. Coin Sorting Accuracy

- **Verification**
- Verify the system correctly separates each coin into its proper denomination bin.
- To be performed by the engineering and testing team
- **Method:** A controlled batch of mixed coins with known quantities will be processed and the results will be compared to the expected sorting outcomes.
- **Equipment needed:** Mixed coin sets, Collection bins, Data sheet
- **Performed during p**rototype testing
- **Ensure that the system meets the requirements of sorting at least 99.5% of coins into correct bins.**

2. Coin Counting Accuracy

- **Verification**
- Verify displayed coin count matches the actual number of coins sorted.
- To be performed by the testing team
- **Method:** The system output will be compared with manual counting of coins after each test run.
- **Equipment needed:** Coin batches, Calculator, Data sheet
- Performed during prototype testing
- Ensure the system displays values that match actual counts with at least 99.5% accuracy.

3. Processing Speed

- **Validation**
- Validate the system meets the required processing speed.
- To be performed by the testing team
- **Method:** The number of coins processed within a set time will be measured.
- **Equipment needed:** Stopwatch, Test coin batch
- Performed during performance evaluation
- Ensure the system processes a minimum of 60 coins per minute.

4. Foreign Object Rejection

- **Verification**
- Verify that non-valid items are properly rejected by the system.
- To be performed by engineering and testing team
- Method: Invalid objects such as washers and foreign coins will be introduced and their output paths will be observed.
- **Equipment needed:** Test objects, Prototype system
- Performed during functional testing
- **Ensure the system rejects at least** 95% of invalid objects correctly.

5. Jam Detection

- **Verification**
- Verify the system can identify and respond to coin jams.
- To be performed by engineering and testing team
- Method: Intentionally block coin path and observe system response.
- **Equipment needed:** Coins, Obstruction setup
- Performed during safety testing
- Ensure the system detects jam and stops operation within 0.5 seconds.

6. Jam Recovery Function

- **Validation**
- Validate the system can attempt to clear jams automatically.
- **To be performed by** testing team
- Method: After creating a jam, the system's recovery sequence will be observed.
- **Equipment needed:** Prototype system, Test coins
- Performed during functional testing
- Ensure the system attempts to clear the jam and either resumes operation or alerts user.

7. Overflow Detection

- **Verification**
- Verify the system detects full bins and redirects coins.
- To be performed by testing team
- Method: Fill bin to capacity and observe system behavior.
- **Equipment needed:** Coins, full bin setup
- Performed during capacity testing
- Ensure the system redirects coins or stops intake when bin is full.

8. Real Time Update Performance

- **Validation**
- Validate the system updated coin counts in real time during operation.
- To be performed by testing team
- Method: Observe display updates while coins are being processed and compare with actual coin flow.

- **Equipment needed:** Prototype system, Test coins
- Performed during integration testing
- Ensure the system displays updates correctly without delay and reflects the actual coin flow,

9. Power Failure Data Retention

- **Verification**
- Verify the system data is preserved during power interruptions.
- To be performed by engineering team
- Method: Power will be cut during operation and system recovery will be evaluated.
- **Equipment needed:** Power supply setup, Prototype system
- Performed during reliability testing
- Ensure coin counts and total values remain unchanged after restart.

10. User Operation Functionality

- **Validation**
- Validate user can operate system functions such as start, stop, reset correctly.
- To be performed by representative users
- **Method:** Users will operate the system without prior instructions while being observed.
- **Equipment needed:** Prototype system
- Performed during user testing phase
- Ensure users operate the system successfully without errors or assistance.